

GEOGRAPHY

PHYSICAL GEOGRAPHY



CLIMATOLOGY

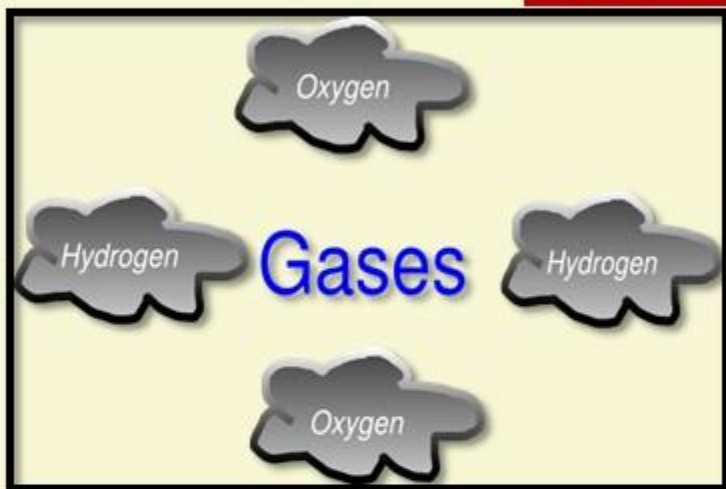
MODULE - 2

COMPOSITION OF ATMOSPHERE

- **PARTICULATE MATTER (PART-1)**

In the Previous Module,

CONSTITUENTS OF ATMOSPHERE



1. GASES



2. WATER VAPOUR

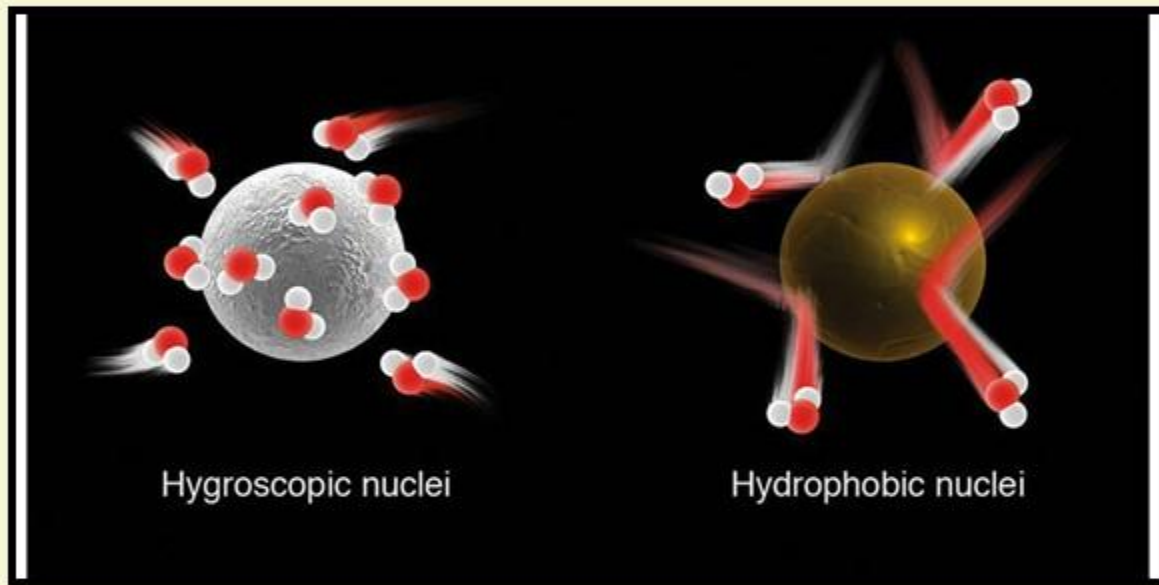


3. PARTICULATE MATTER



CONSTITUENTS OF ATMOSPHERE

PARTICULATE MATTER



WHAT IS HYGROSCOPIC NUCLEI?

A microscopic particle in the free air on to which water vapour may condense to form droplets.

WHAT IS PARTICULATE MATTER?

- PM or SPM(Suspended Particulate Matter) are **microscopic solid or liquid matter suspended in Earth's atmosphere**.
- Comprises of dust, salt, pollen, smoke, soot, volcanic ashes etc.
- Helps in **scattering of solar radiation** which adds charming colour of red and orange at sunrise and sunset.
- These particles becomes **hygroscopic nuclei** and thus helps in precipitation.

CONSTITUENTS OF ATMOSPHERE

PARTICULATE MATTER



VOLCANIC ASHES



HAZE



POLLEN



SMOKE



SOOT

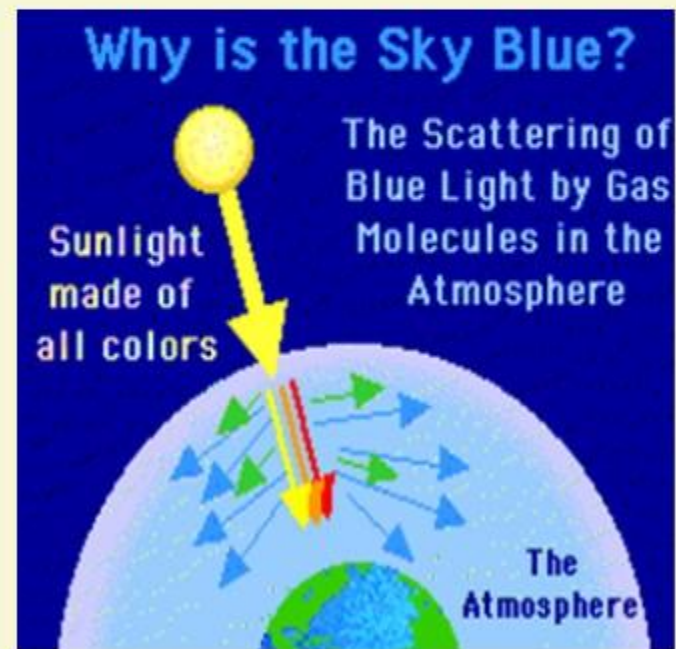
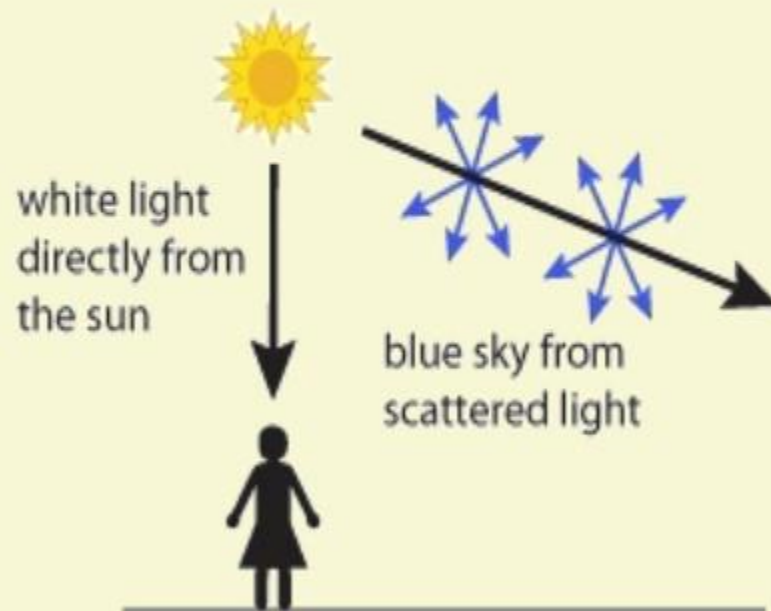


DUST

CONSTITUENTS OF ATMOSPHERE

PARTICULATE MATTER

SCATTERING OF LIGHT



CONSTITUENTS OF ATMOSPHERE

PARTICULATE MATTER

WHAT PM IS NOT?

BEWARE THESE ARE NOT PM!!

NITROGEN DIOXIDE



OZONE



- Finally, it's helpful to think of what PM is not.
- Particulate pollution **does not** include gas pollutants like ozone and NO_2 !!

CONSTITUENTS OF ATMOSPHERE

PARTICULATE MATTER

SOURCES OF PARTICULATE MATTER



DUST STORMS



SEA SPRAY



VOLCANOES



FOREST FIRES

NATURAL SOURCES

- Volcanoes
- dust storms
- forest and grassland fires
- vegetation
- sea spray

CONSTITUENTS OF ATMOSPHERE

PARTICULATE MATTER

SOURCES OF PARTICULATE MATTER



VEHICLE EXHAUST



INCINERATORS



POWER PLANTS



**CONSTRUCTION
SITES**

ANTHROPOGENIC SOURCES

- Burning of fossil fuels in vehicles
- power plants and various industrial processes
- incinerators
- construction sites

CONSTITUENTS OF ATMOSPHERE

PARTICULATE MATTER

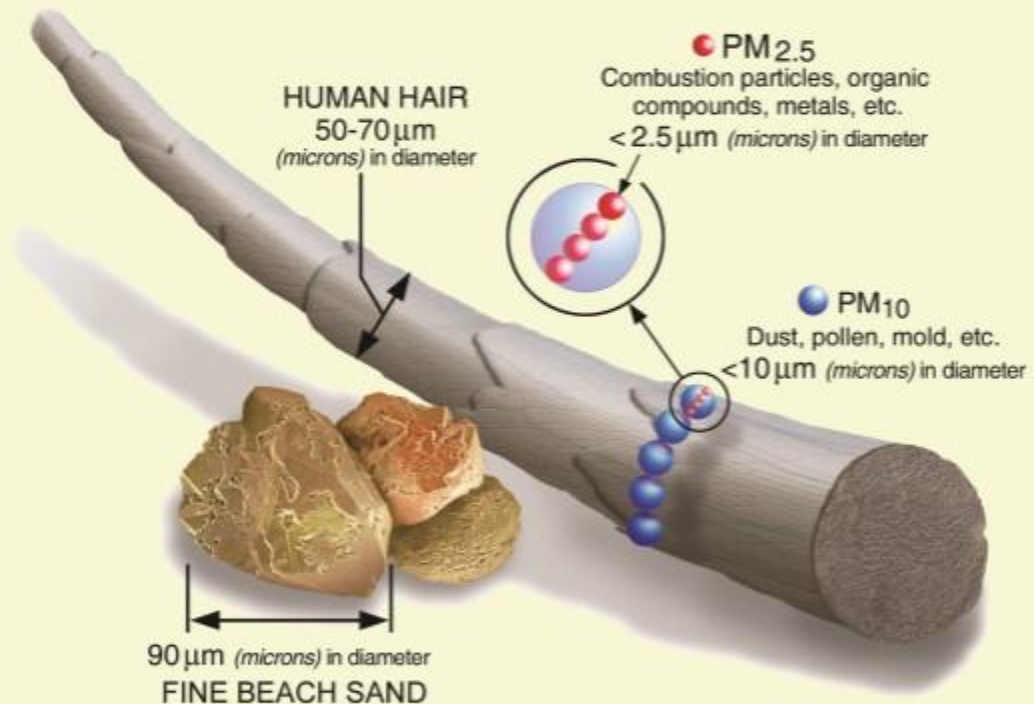
SIZES OF PARTICULATE MATTER

1 PM₁₀

Coarser particles; Inhalable particles, with diameters that are 10 micrometers and below (average human hair is about 70 micrometers in diameter).

2 PM_{2.5}

Fine inhalable particles; 2.5 micrometers and smaller.



CONSTITUENTS OF ATMOSPHERE

PARTICULATE MATTER

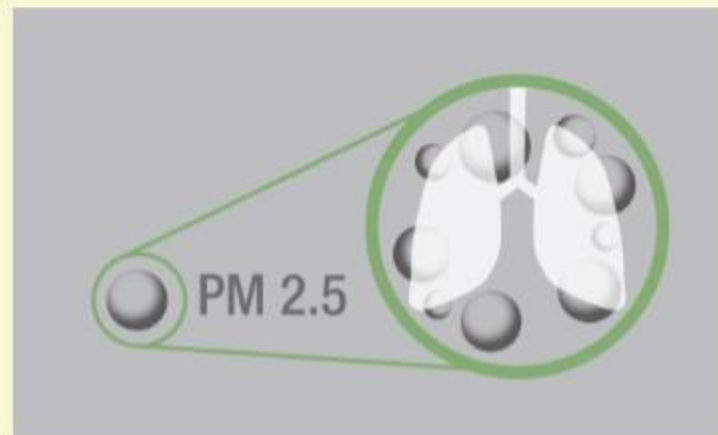
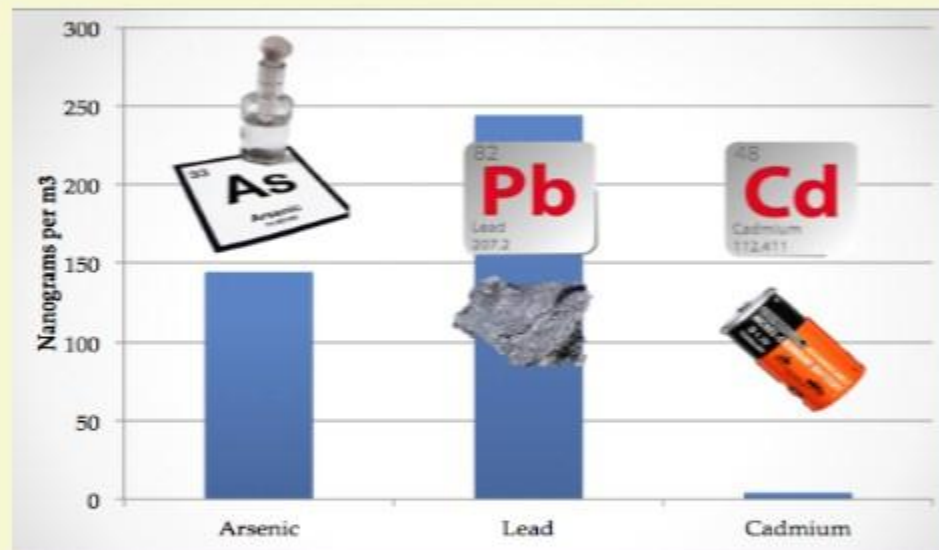
HARMFUL EFFECTS OF PARTICULATE MATTER

1 PM2.5 is a part and parcel of PM10.

2 PM2.5 is much more harmful than PM10.

WHY IS IT SO?

While PM10's story ends at the lungs, PM2.5 can pass from our lungs into our blood supply and be carried throughout our bodies thereby making them "the invisible killer".



CONSTITUENTS OF ATMOSPHERE

PARTICULATE MATTER

STANDARD VALUES OF PARTICULATE MATTER

According to the **National Ambient Air Quality Standards(NAAQS)** of the **Central Pollution Control Board (CPCB)** of India, the 24-hour average for

- **PM10** is **100** microgram/cubic metre
- **PM2.5** is **60** microgram/ cubic metre.

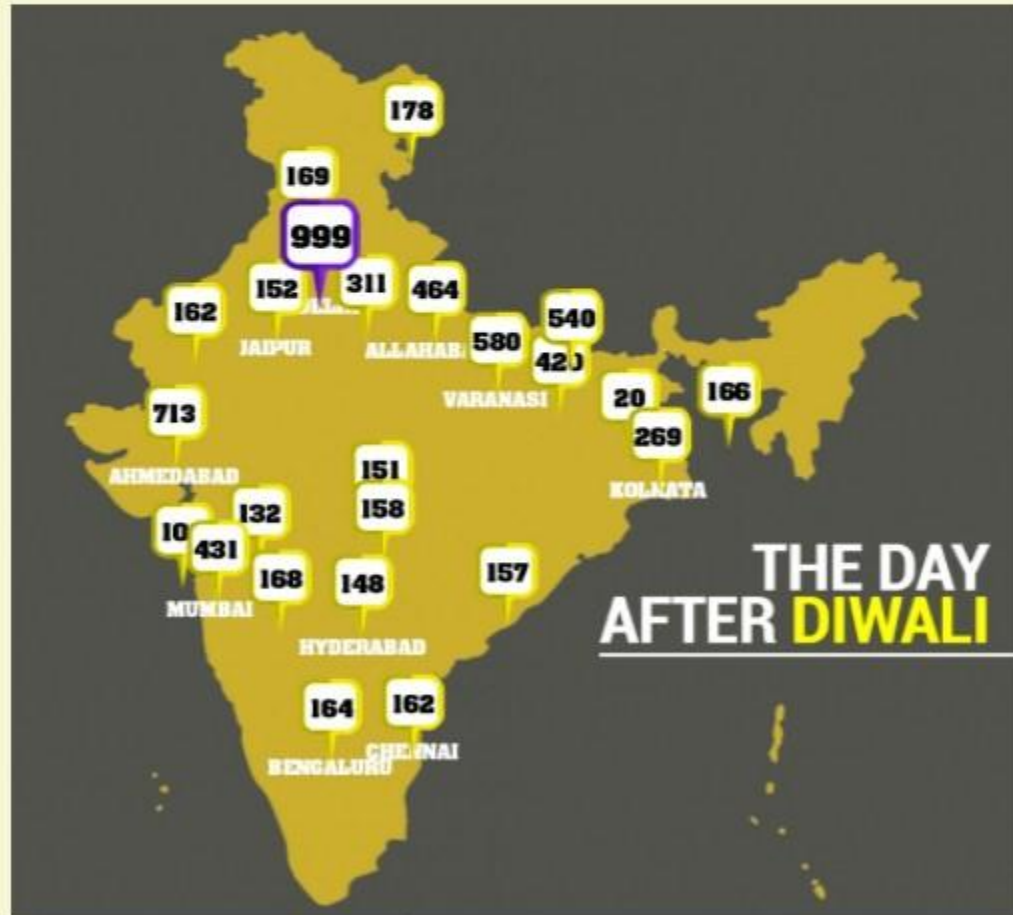


CONSTITUENTS OF ATMOSPHERE

PARTICULATE MATTER

HARMFUL EFFECTS OF PARTICULATE MATTER

AFTER DIWALI, PM_{2.5} WAS 16 TIMES MORE THAN STANDARD VALUES!!



During the **Delhi smog**, PM_{2.5} reached as high as 999 microgram/cubic metre!!



CONSTITUENTS OF ATMOSPHERE

PARTICULATE MATTER

POSSIBLE MAINS QUESTION

Topic: GS Paper 1 : Urbanization- their problems and remedies

GS Paper 3 : Environmental pollution and degradation

- 1** What is the difference between PM2.5 and PM10 with respect to the atmospheric pollution?(5 Marks) (100 words)
- 2** Examine the health impacts of PM pollution in India. Also critically comment on government's response to control such pollution. (12 Marks)(250 words)

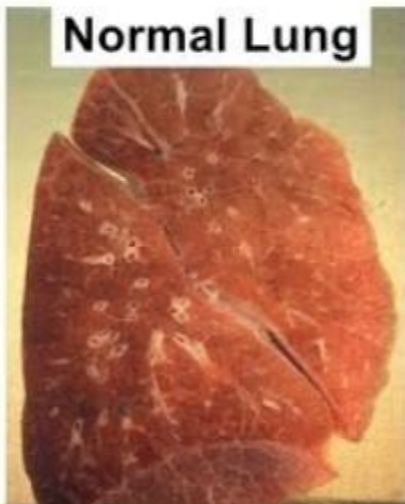
CONSTITUENTS OF ATMOSPHERE

PARTICULATE MATTER

HARMFUL EFFECTS OF PARTICULATE MATTER

Particulate Matter: Health Effects

Normal Lung

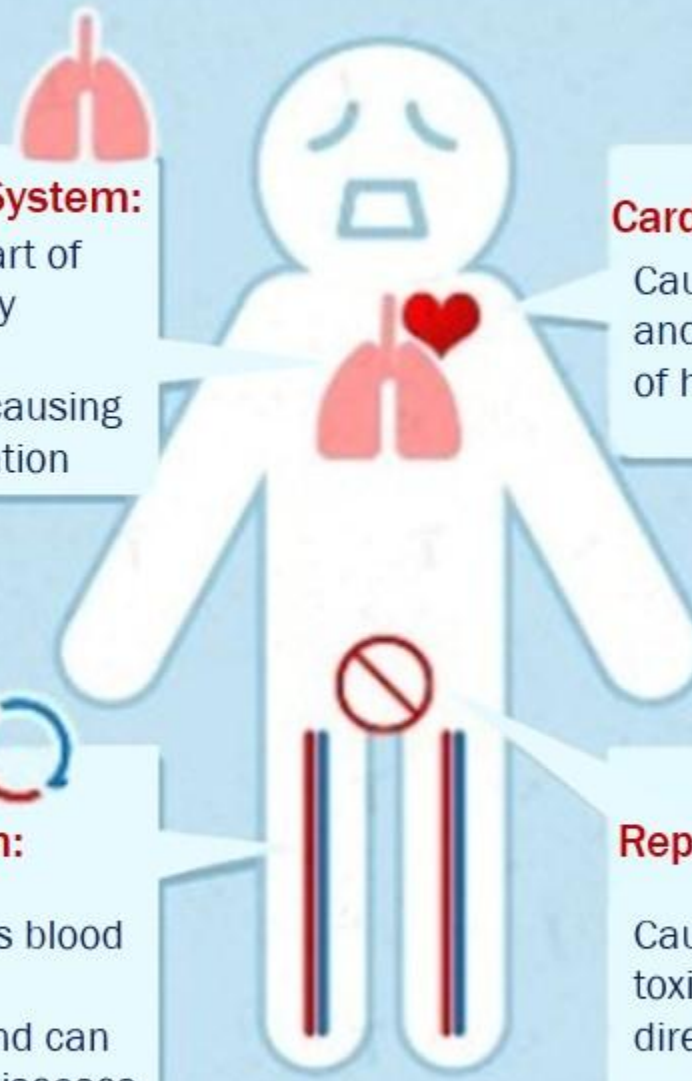


Polluted Lung



- 1 Acute or short-term & sudden exposure** can cause chest tightening, watery eyes, sneezing, and running nose.
- 2 Chronic or long-term exposure** can cause asthma, pulmonary diseases and any type of bronchitis.
- 3** PM 2.5 is declared as Class-1 Cancer-causing agent (carcinogen) in 2013 by the **International Agency for Research on Cancer(IARC)**, which is part of the WHO.

HARMFUL EFFECTS OF PARTICULATE MATTER



Respiratory System:

The soluble part of PM 2.5 directly enters the bloodstream causing lung inflammation

Cardiovascular System:

Causes cardio toxicity and disrupts the activity of heart muscle.

Blood System:

PM 2.5 causes blood toxicity, blood coagulation and can trigger heart diseases

Reproductive System:

Causes placental toxicity and leads to direct harm to fetus.

Groups sensitive to PM_{2.5}

People with heart or respiratory diseases, the elderly, pregnant women and children



CONSTITUENTS OF ATMOSPHERE

PARTICULATE MATTER

HEAVY METAL POLLUTION

REAL DANGER LIES IN HEAVY METALS!!



1 Heavy metals are defined as metallic elements that have a relatively high density compared to water.

2 When present above permissible limits, it can cause DNA damage that may lead to cell cycle modulation and carcinogenesis.

***DO GIVE A READ ON HEAVY
METAL POISONING!!***

CONSTITUENTS OF ATMOSPHERE

PARTICULATE MATTER

HEAVY METAL POLLUTION

2. Which of the following can be found as pollutants in the drinking water in some parts of India? (UPSC 2013)

1. Arsenic
2. Sorbitol
3. Fluoride
4. Formaldehyde
5. Uranium

Select the correct answer using the codes given below.

- a) 1 and 3 only
- b) 2, 4 and 5 only
- c) 1, 3 and 5 only**
- d) 1, 2, 3, 4 and 5

PARTICULATE MATTER

HEAVY METAL POLLUTION

FLUORIDE AND ARSENIC ARE “FAMOUS” WATER POLLUTANTS!!

Skeletal and Dental Fluorosis



Skin pigmentation/ keratosis of Arsenic



2. Which of the following can be found as pollutants in the drinking water in some parts of India? (UPSC 2013)

1. Arsenic
2. Sorbitol
3. Fluoride
4. Formaldehyde
5. Uranium

Select the correct answer using the codes given below.

- a) 1 and 3 only
- b) 2, 4 and 5 only
- c) 1, 3 and 5 only**
- d) 1, 2, 3, 4 and 5

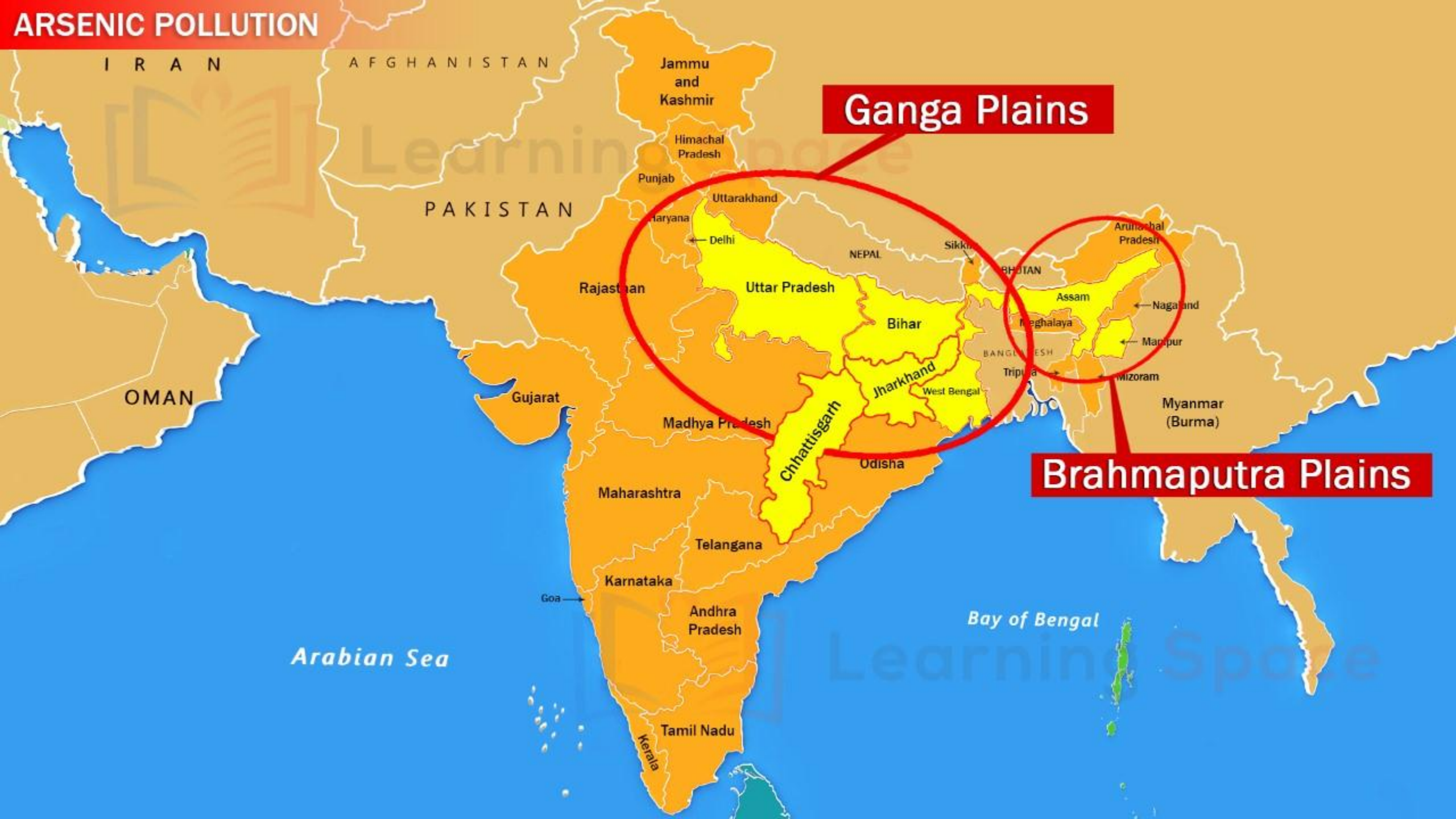
HOW DOES THESE POLLUTANTS FORM?

- Underground water exploitation
- Industrial pollution
- Excessive usage of fertilisers
- Unchecked mining

FLUORIDE POLLUTION



ARSENIC POLLUTION



CONSTITUENTS OF ATMOSPHERE

PARTICULATE MATTER

HEAVY METAL POLLUTION

2. Which of the following can be found as pollutants in the drinking water in some parts of India? (UPSC 2013)

1. Arsenic
2. Sorbitol
3. Fluoride
4. Formaldehyde
5. Uranium

Select the correct answer using the codes given below.

- a) 1 & 3 only b) 2, 4 & 5 only **c) 1, 3 & 5 only** d) 1, 2, 3, 4 & 5

SORBITOL AND FORMALDEHYDE

- Sorbitol- It is a sweetener and also used for constipation problems.
- Formaldehyde is used in coating, laminating; also used in **preservation of animal species.**

Formaldehyde is an Indoor air pollutant (Sick Building Syndrome)!!



Foods High in Sorbitol



CONSTITUENTS OF ATMOSPHERE

PARTICULATE MATTER

HEAVY METAL POLLUTION

2. Which of the following can be found as pollutants in the drinking water in some parts of India? (UPSC 2013)

1. Arsenic
2. Sorbitol
3. Fluoride
4. Formaldehyde
5. Uranium

Select the correct answer using the codes given below.

- a) 1 & 3 only b) 2, 4 & 5 only **c) 1, 3 & 5 only** d) 1, 2, 3, 4 & 5

URANIUM POLLUTION IN PUNJAB

Uranium in ground water is indeed a problem in Punjab leading to birth of mentally retarded children



CONSTITUENTS OF ATMOSPHERE

PARTICULATE MATTER

POSSIBLE MAINS QUESTION

Topic: GS Paper 1 : Urbanization- their problems and remedies

GS Paper 3 : Environmental pollution and degradation

- 1** What is the difference between PM_{2.5} and PM₁₀ with respect to the atmospheric pollution?(5 Marks) (100 words)
- 2** Examine the health impacts of PM pollution in India. **Also critically comment on government's response to control such pollution.** (12 Marks)(250 words)

WE EXPRESS OUR SINCERE THANKS FOR VIEWING THIS VIDEO

Presented by



Learning Space

For Suggestions:

suggestions@learningspace.in

To Contact us:

info@learningspace.in

OUR TEAM

G. V. Rao

K. Srikanth

D. Sunil Kumar

L. V. Krishna

D. Chaitanya

Amrita Naidu

M. Binukrishna

G. Ravi Babu

D. Joji

K. Victor Babu

Visit us at:

www.learningspacedigital.com

0 8 6 6 - 2 4 4 4 7 2

0 9 8 4 9 9 4 2 2 9 9